

Experiment Report

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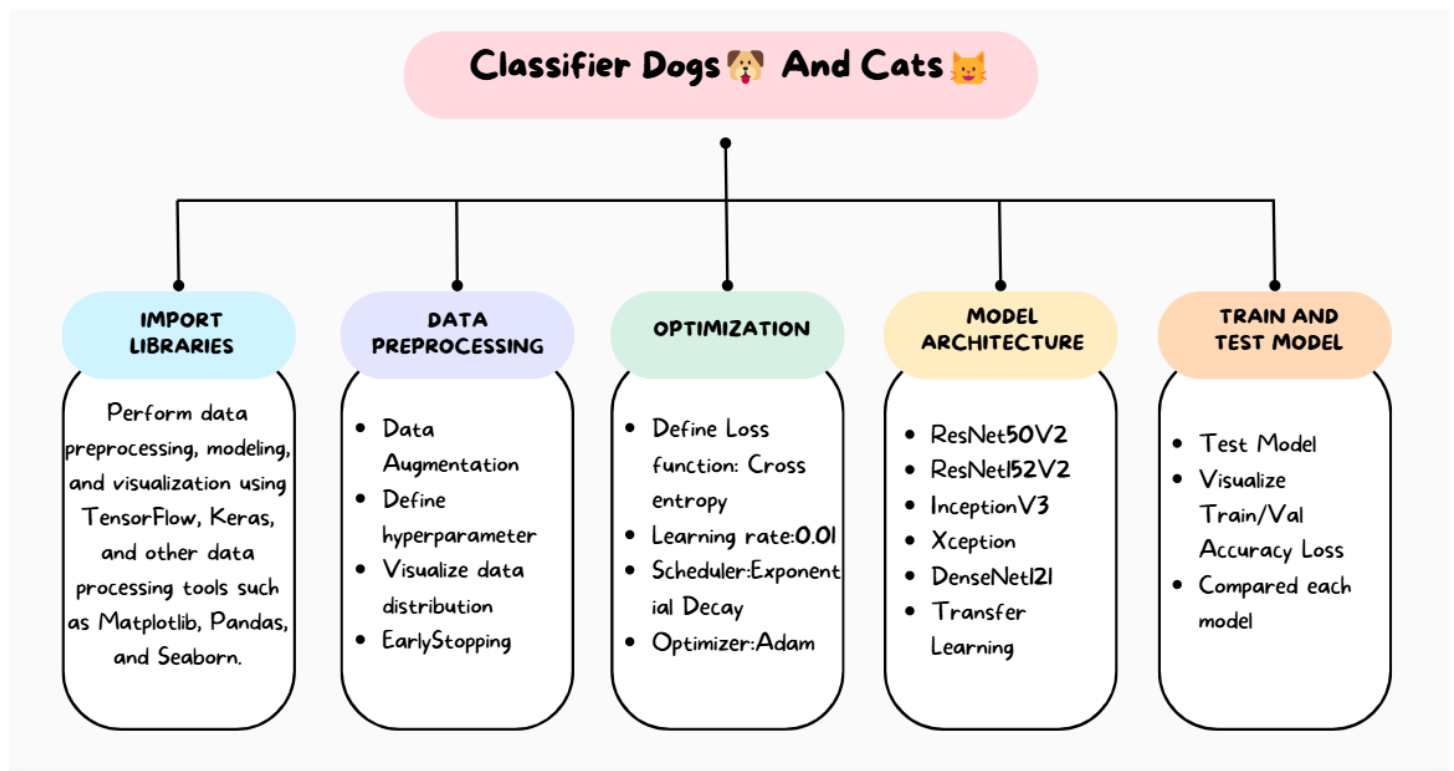
Professor: Ching-Wen Ma

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Abstract

This assignment involves studying a Kaggle competition to understand the complete workflow of model training. The goal is to gain insights into methodologies used by other practitioners and explore essential steps for model optimization. Key tasks include constructing a flowchart to represent the experimental process and identifying methods or factors crucial to enhancing model accuracy.

Flowchart



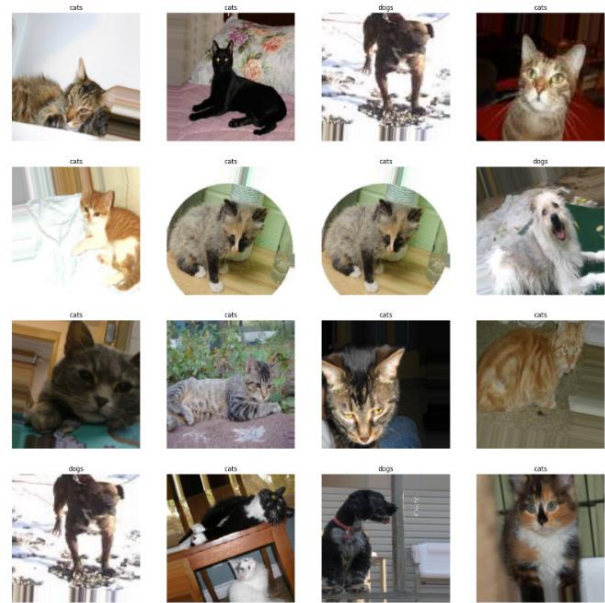
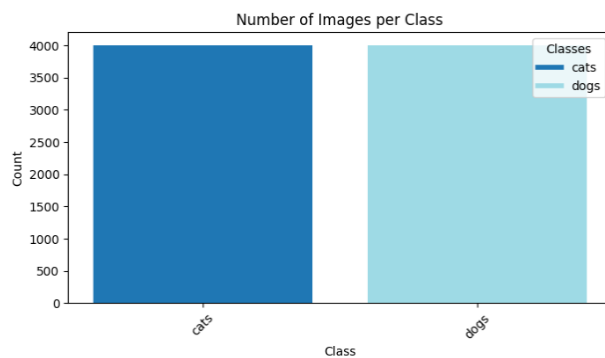
Step1. Import some necessary libraries to train, visualize and so on task.

Step2. Data Preprocessing:

- Data Augmentation
 1. Random Rotation: Images were rotated randomly within a 20-degree range.
 2. Horizontal and Vertical Shifts: The images underwent horizontal and vertical shifts by 20% of the image dimensions.
 3. Random Horizontal Flipping: Each image was flipped horizontally at random.
 4. Fill Mode: Pixels introduced due to transformations were filled using a nearest-neighbor method.
- Hyperparameter Definition

1. Image Size: Set to 200 pixels to balance detail preservation with computational efficiency.
2. Batch Size: Set to 32 to optimize training speed and model convergence.
3. Validation Split: 20% of the dataset was allocated for validation to monitor model generalization.
4. Epochs: Training was limited to 10 epochs, given the data size and expected convergence rate.

- Data Distribution Visualization



- Early Stopping: Early Stopping was implemented to monitor the validation set's performance as an indicator of overfitting. Training was halted once validation performance plateaued or began declining, thus ensuring the model did not excessively fit the training data.

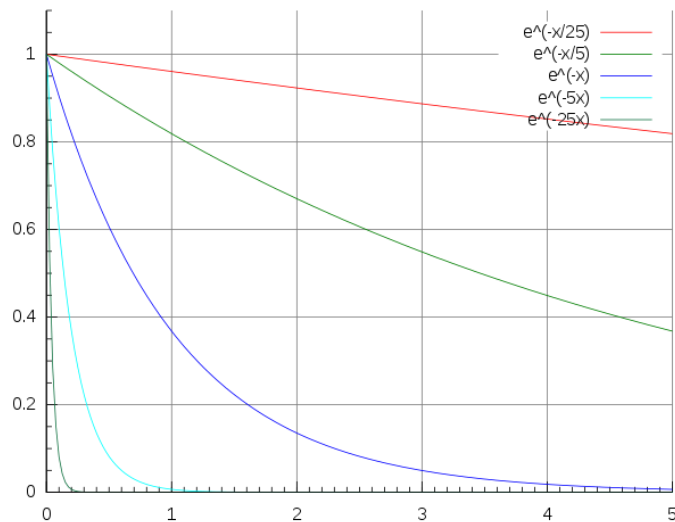
Step3.Optimization

- Define Loss function: Cross Entropy

$$H = \sum_{c=1}^C \sum_{i=1}^n -y_{c,i} \log_2(p_{c,i})$$

- Learning rate: 0.01

- Scheduler: Exponential Decay



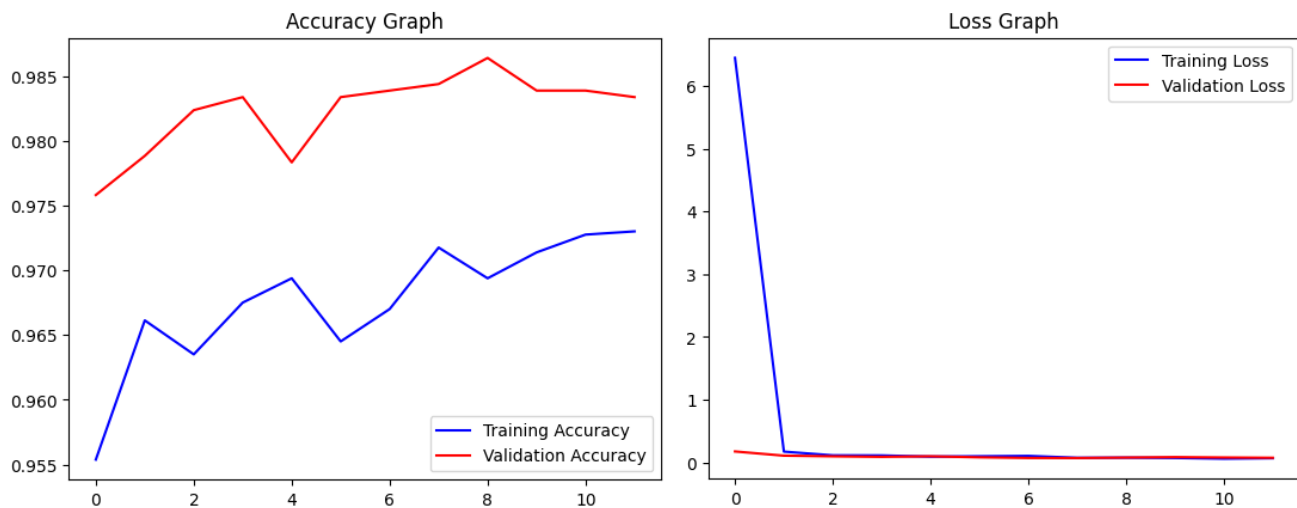
- Optimizer: Adam

Step4.Model Architecture

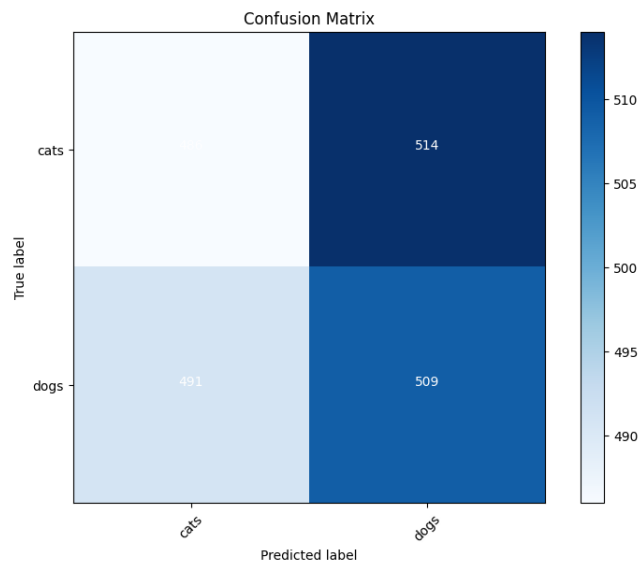
- ResNet50V2
- ResNet152V2
- InceptionV3
- Xception
- DenseNet121
- Transfer Learning: Xception to finetune

Step5.Train and Test model

- Visualize training/validation epoch



- Confusion matrix



- Output the test data with predicted label by model



Conclusion

This assignment delved into the Kaggle competition process, highlighting key steps in model training and optimization. Our flowchart detailed essential techniques from data preprocessing to evaluation, underscoring the significance of data augmentation, hyperparameter optimization, and the use of advanced model architectures. Implementing methods like early stopping and exploring diverse architectures such as ResNet and Xception emphasized practical strategies for

improving model robustness and accuracy. This study equips us with foundational strategies for enhancing performance in competitive machine learning settings.

References

1. Mohammad Alaliwi (2023). Classifier dogs 🐶 or cats 🐱 (val_acc=+98.5%) Retrieved from <https://www.kaggle.com/code/mhmmadalewi/classifier-dogs-or-cats-val-acc-98-5>